# Naghmeh (Melody) Nazar

**FALL 2027 PhD APPLICANT | SPEECH ML | TRUSTWORTHY AI & AGENT SECURITY INTERESTS**

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## RESEARCH FOCUS

Fall 2027 PhD applicant with completed M.S. research in speech ML and developing interests in trustworthy AI and tool-using agent security. Independently studying selected attack patterns documented in MCP-SafetyBench - external work by Zong et al. (ICLR 2026) - and considering how pre-execution safeguards could be evaluated. This exploration has no benchmark authorship, implemented defense, or validated results. M.S. thesis: four-class speech emotion recognition on RAVDESS and TESS, with explicit limits on speaker-independent and cross-corpus claims. Brings 3+ years of production software engineering experience to building testable AI systems.

**Research interests:** LLM agent security, AI safety, indirect prompt injection, scalable oversight, adversarial evaluation, speech processing, affective computing

## RESEARCH EXPERIENCE & INDEPENDENT EXPLORATION

**Independent Exploration - Safety of Tool-Using LLM Agents** **Ongoing**

- Studying selected attack patterns documented in MCP-SafetyBench (Zong et al., ICLR 2026), an external benchmark, to understand how untrusted context can affect tool selection and arguments. External reference: MCP-SafetyBench paper
- Considering how a pre-execution risk check could be evaluated using attack success, benign task completion, false-block rate, and calibration; no experiments, defense implementation, or validated results are claimed.
- Built an n8n email-and-scheduling agent and documented where untrusted messages and generated actions require validation or human approval. github.com/nzrnaghme/RespondEmailAgent

**M.S. Thesis - Speech Emotion Recognition, CSUN** **Aug 2026**

- Developed a four-class 2D ResNet-inspired CNN on RAVDESS and TESS using 16 kHz audio and 128 x 128 log-Mel spectrograms.
- On the current mixed-corpus split, achieved 80.36% validation accuracy and 74.4% held-out test accuracy; the split does not establish speaker-independent or cross-corpus generalization.
- Desktop latency and naturalistic-speech generalization remain unmeasured; no deployment or clinical-validation claim is made.

## RESEARCH OUTPUTS & PRESENTATION

- M.S. thesis - "Real-Time Speech Emotion Recognition Using Optimized Deep Residual Networks," California State University, Northridge, Aug 2026. PDF available on request.
- Presenter - 40th Annual CSU Student Research Competition: M.S. speech emotion recognition research, 2026. Presentation materials available on request.

## EDUCATION

**M.S. in Computer Engineering, California State University, Northridge** **Aug 2024 - Aug 2026**
Thesis advisor: Prof. Shahnam Mirzaei

**B.S. in Computer Engineering, University of Science and Technology of Mazandaran** **Aug 2016 - Mar 2021**

## PROFESSIONAL EXPERIENCE

**Senior Frontend Developer, Golrang Industrial Group - Tehran, Iran** **Jul 2022 - Jul 2024**

- Architected cross-platform applications with React Native and React.js PWA; contributed UI implementation, state management, caching, performance optimization, and code review.

**React Developer, Erole.ir - Tehran, Iran** **Jun 2021 - Jun 2022**

- Built React and Redux applications and integrated REST APIs and SignalR-based real-time features.

## TECHNICAL SKILLS

**Topics explored independently:** indirect prompt injection, tool-call risk, human-approval boundaries

**Speech and machine learning:** Python, TensorFlow, Keras, scikit-learn, NumPy, Pandas, 2D CNN/ResNet, log-Mel spectrograms, MFCCs

**Engineering:** C++, JavaScript, TypeScript, n8n, React, React Native, Node.js, REST APIs, Git, Linux